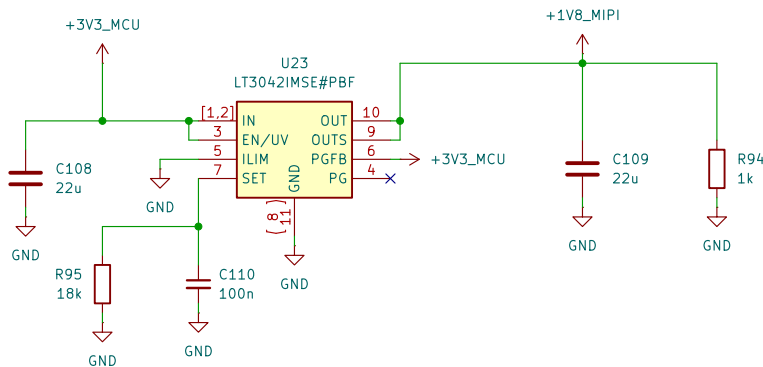
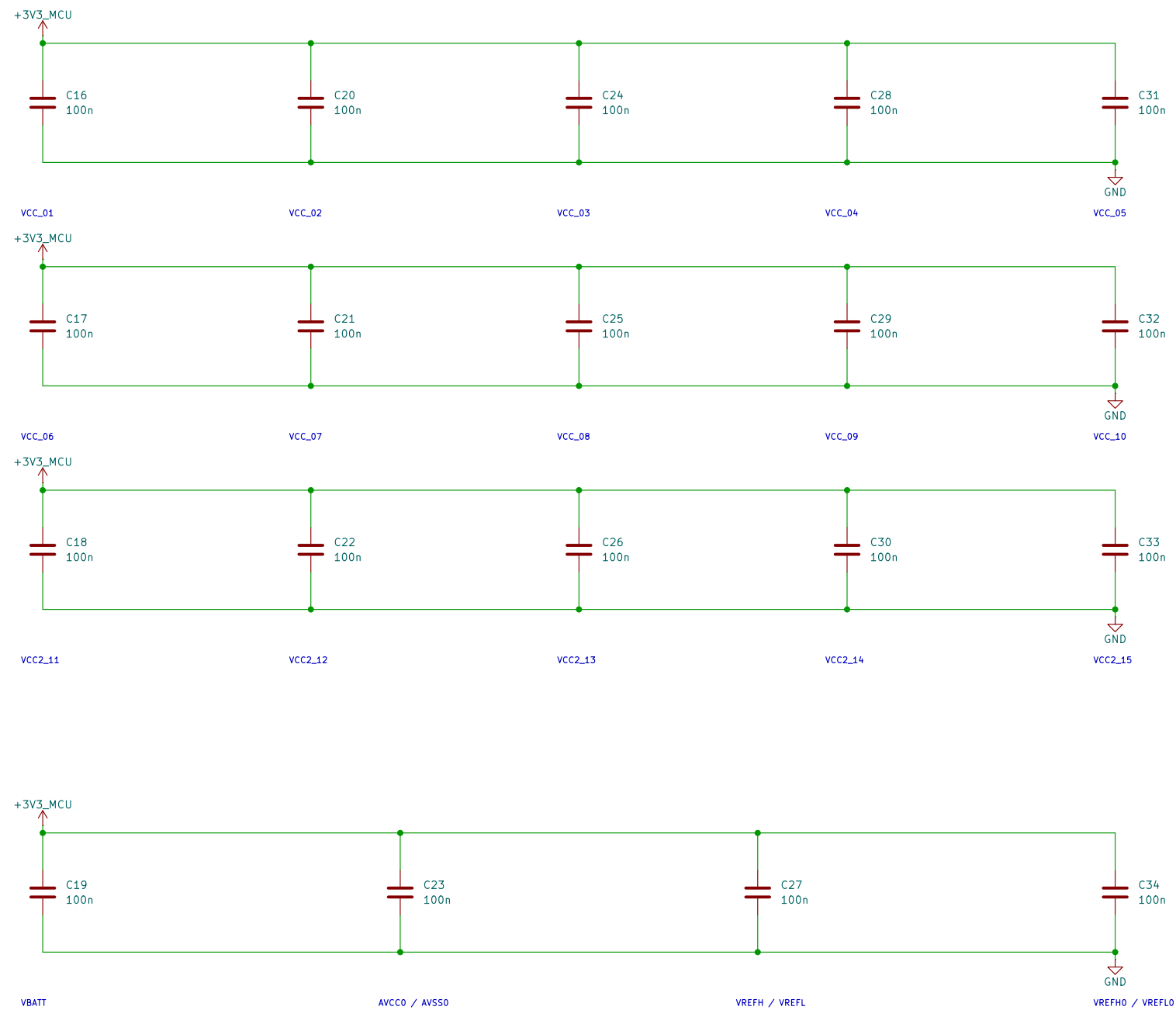


[illegible]

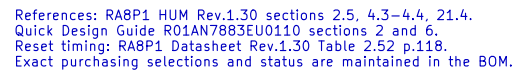
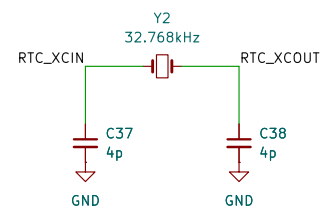
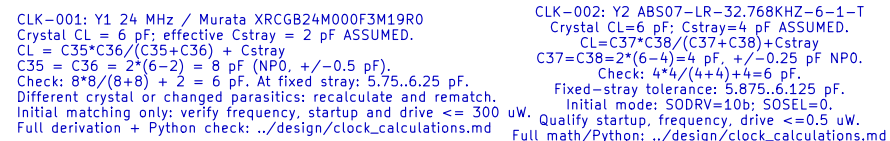


I/O DECOUPLING – 100 nF AT EACH NUMBERED VCC / VSS PAIR

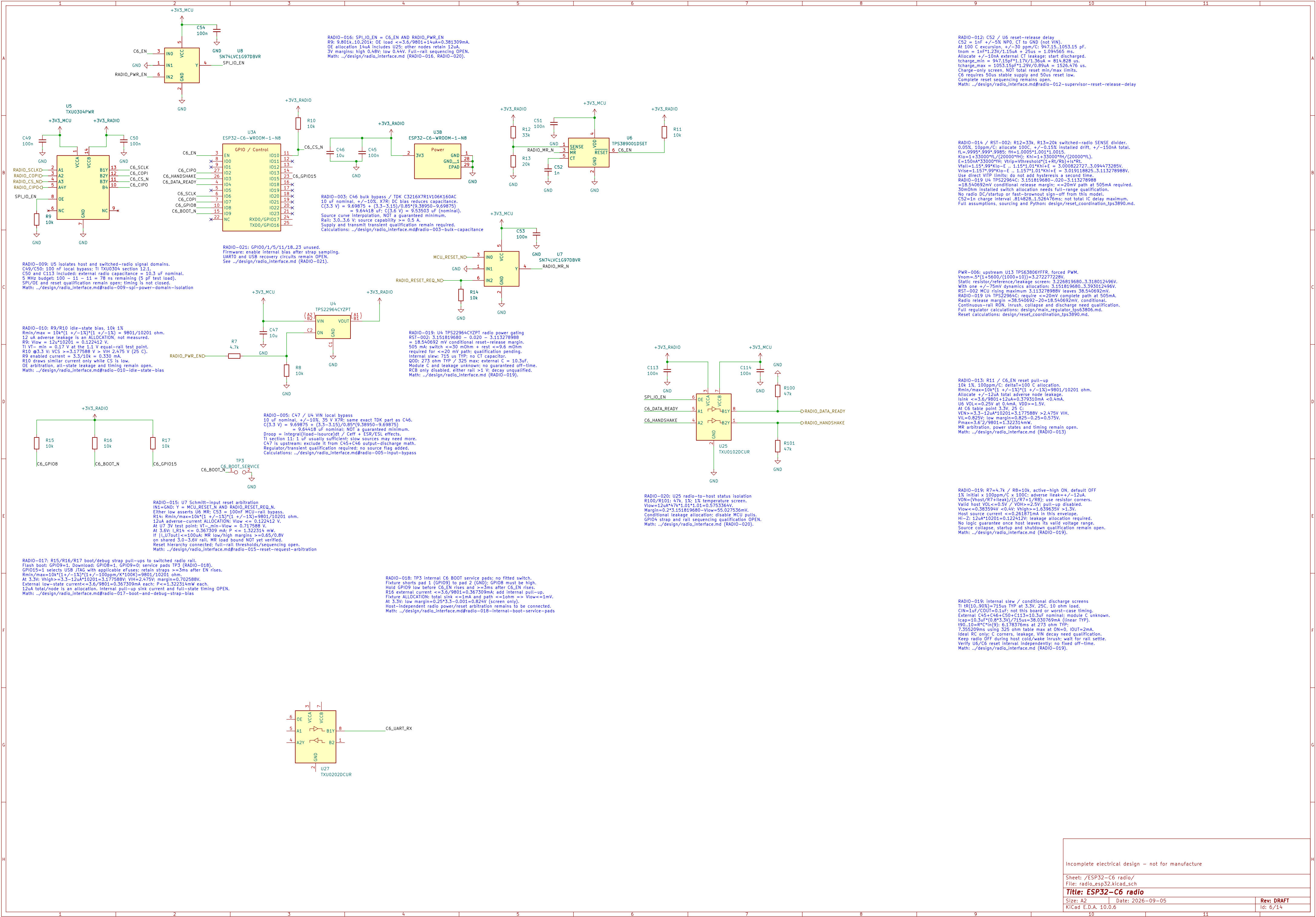


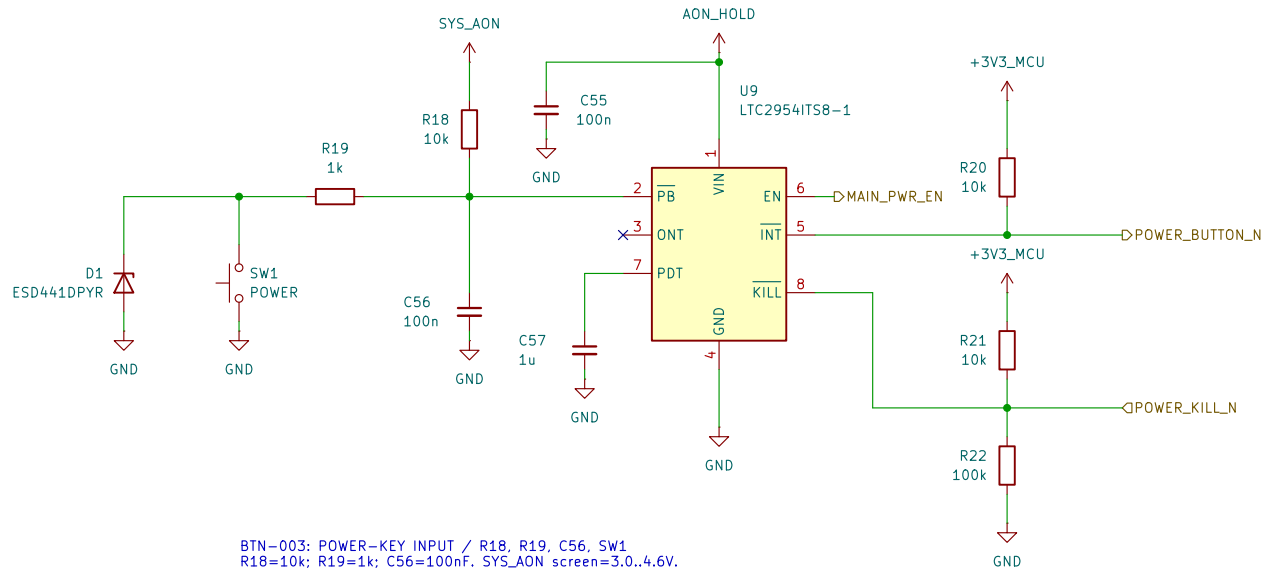
CMS-013: +20mA steady CSI branch planning allocation
4.1mA PHY + 1.989593mA bleed + 13mA allocated GND
+ 0.102mA SET + 0.100mA auxiliary = 19.291593mA.
13mA is an engineering envelope, not an interpolated guarantee.
Auxiliary allowance requires verification: startup/inrush excluded.
AVCC_MIP1 remains within existing MCU allocation.
Main subtotal: 2.075 + 0.020 = 2.095mA; full system budget pending.
At 2.0496V, 3.2V input, 1.25W average, 75% efficiency:
input 2.961817158A; converter loss = 2.3694535727W
Source, current and thermal acceptance remain open.
Sources, conditions and Python: design/camera_storage/interfaces.md.

RA8P1 e-reader		
Sheet: /RA8P1 IO and analog supplies/ File: mcu_supply_power.kicad_sch		
Title: RA8P1 I/O and analog supplies		
Size: A3	Date: 2026-09-05	Rev: 0.3 DRAFT
KiCad E.D.A. 10.0.6		Id: 4/14



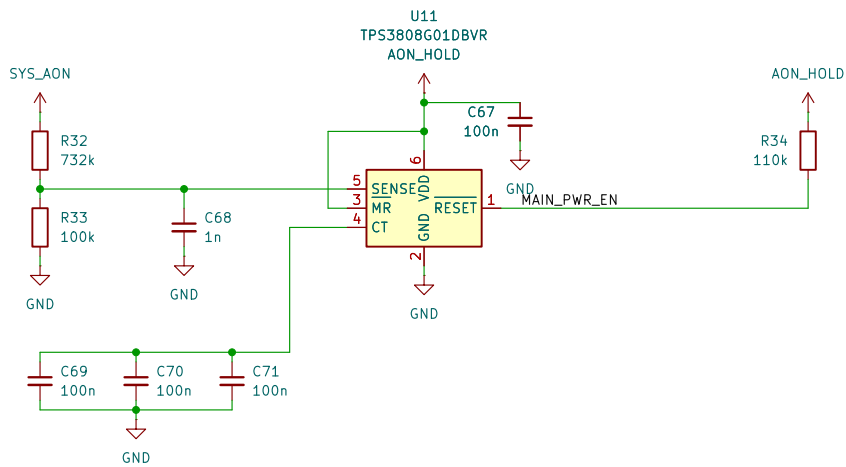
```
RST-002: U2 TPS3890U1DSET; R67=33k, R74=20k, R75=10k, C95=10n.  
Sense resistors: +0.05%, OhmCo over 100C, allocated + / -0.15% assembly/aging.  
Klo=9995.99*Klo-E = 9.9902E+25; E=1.0005E+10; Klo=330000.00*Fh; Fh=330000.00*Fh;  
Klo=4.133000e/f/(200000*f.h); E=1.33000e/f/(20000*f.h); E=150nA/330000f.h.  
Vfall=1.15*.99*Klo-E = 1.15*1.01*Khi+E = 3.00082727Z.3.094473285V.  
Vrise=1.157*.99*Klo-E = 1.157*1.01*Khi+E = 3.01911882S.3.113278988V.  
Main release margin=3.151819680-.3.113278988=.38.540692mV; no double hysteresis.  
C95: 10n COG +/- .5%, 30ppm/C over 100C; allocate + / -10nA external CT leakage.  
tcharge,min=10n*.95*.997*1.17/(1.35uA=10n).E148276ms> 2.4ms RES minimum.  
tcharge,max=10n*1.05*1.003*1.29/(90u=10n)=15.264758ms; NOT total-delay max.  
R75 MR pull-up: R=9801.10201ohm; allocate + / -50uA total adverse MR current.  
MR high margin=3.3*1.51819680-50uA/10201=.435459504V; sink <- .396190439mA.  
Conditional DC/startup design, not fast-collapse or SDRAM-quiescence proof.  
Full derivation, assumptions, sourcing and Python: design/reset_coordination_tps3890.md.
```





BTN-003: POWER-KEY INPUT / R18, R19, C56, SW1
R18=10k; R19=1k; C56=100nF. SYS_AON screen=3.0..4.6V.
R bounds: nominal*(0.99*0.99)..nominal*(1.01*1.01).
At PB=0.6V: Isource<=(4.6-0.6)/9801 + 15uA + 12uA
<435.13uA; Isink=0.6/(1020.1+0.5)>587.88uA.
Thus the closed switch pulls PB below its 0.6V falling threshold.
Min contact current=3/(10201+1020.1+0.5)-12uA>255.34uA.
Open contact>=2.87748V; SW1 minimum rating=10uA at 2V.
C56 discharge peak<=4.6/980.1<4.694mA < SW1 50mA rating.
12uA is a leakage ALLOCATION; ESD/transient validation required.
Full derivation + Python: ../design/single_button_power.md BTN-003/005/010.

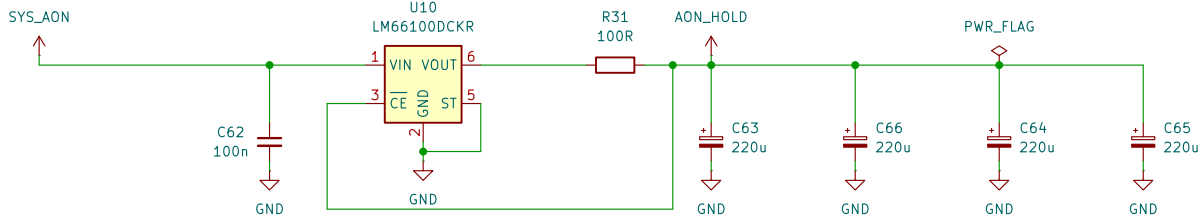
BTN-004: HARDWARE FORCED-OFF TIMING / C57
C57=1uF; TDK C2012X7R1E105K125AB, 25V X7R +/-10%.
C[uF]=1.56e-4*(t_added[ms]-1).
t_force,nom=64ms + 1ms + 1/1.56e-4=6475.256ms (6.48s).
C tolerance/temp only: 1*0.90*0.85=0.765uF.
1*1.10*1.15=1.265uF -> 4.969..8.174s using NOMINAL IC equation.
Not guaranteed timing: excludes IC spread, DC bias, aging, leakage.
ONT intentionally open: internal turn-on debounce=26..41ms.
Release the initial ON press; after forced off, release then press again.
Full calculation + Python: ../design/single_button_power.md BTN-004/005.



SYS-007 / SENS-010: SHUTDOWN / U12, Q1-Q3, R35-R40, R70-R71
POWER_OFF_H=NOT(MAIN_PWR_EN); held-domain VCC=2.7..4.6V.
Each gate branch: 1k series / 1M pulldown; 1% +100ppm/C, 100C.
4-gate peak<=4*4.6/(1000*0.99^2)=18.773595mA <50mA.
Includes future USB-clear; static<=22.773595uA <100uA.
VGSmin=(2.6-1020.1*1uA)/(1+1020.1/980100)=2.596278V.
1uA per gate/board leakage (is a qualification allocation).
RUN/KILL/discharge/USB-clear FET drains remain separate.
R39||R40 + Q2: Rmax=22*1.01^2/2+0.2=11.4211ohm.
Irc<=2.5mA: 1.5mA key-filter return +1mA other injection.
Vinf=Irc*Rmax=28.552750mV; Cmain<=1mF.
Total=10ms+Rmax*Cmain*ln((3.6-Vinf)/(0.09-Vinf))+10ms
=66.398757ms <75.794309ms hold; margin9.395552ms.
Each 22R: P<=3.6^2/(22*0.99^2)=0.601052W; 1W at70C.
Qualify FET resistance, response, derating, backfeed and cycling.
Gate/transient charge shares 1uC reserve; allocations need validation.
VBATT firmware/option invariant remains unimplemented.
Sources/Python: ../design/sensor_alternative.md SENS-010;
../design/system_power_design.md SYS-007 for topology.

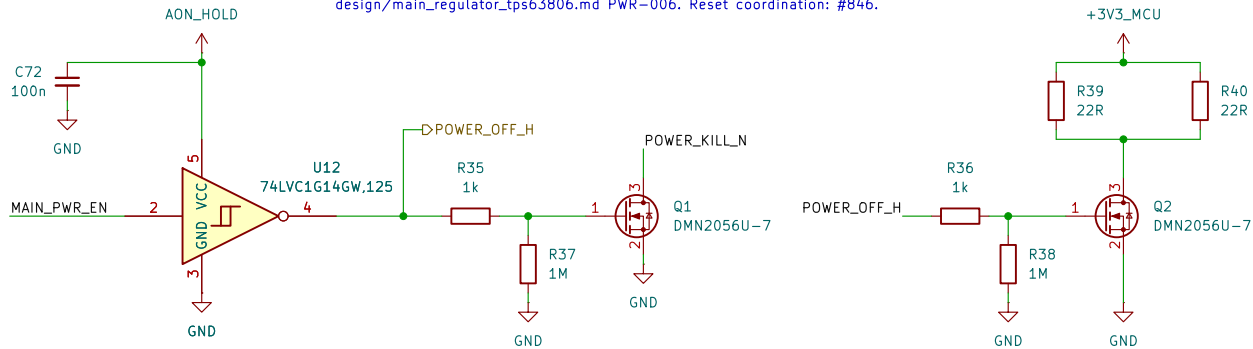
INCOMPLETE POWER CONTROL - NOT FOR FABRICATION
PB, INT, KILL, held supply and source supervisor are connected.
U12/Q1/Q2 provide fault KILL clamp and main-rail discharge.
MAIN_PWR_EN feeds raw U16; held Q3 separately clamps U13 EN.
SYS_AON source and USB permission-latch clear remain to be built.
Current converter: U13 TPS63806; design/main_regulator_tps63806.md PWR-006.
MCU reset: design/reset_coordination_tps3890.md RST-002.
Radio release margin and system shutdown coordination remain open (#846).

ERC supply path: U10 VOUT through R31.
PWR_FLAG declares that series-fed source only;
raw SYS_AON source remains unimplemented.



SYS-007 / SENS-010: HELD SUPPLY / U10, R31, C62-C66
C63-C66: 4 x 220uF/10V; Cmin=880uF*0.9^3=641.52uF.
VIN=raw SYS; VOUT->R31->reservoir; CE senses reservoir.
Icontrol<=1.825mA incl 1.100mA bank leakage screen.
Reverse allowance=0.817mA; Itotal<=2.642mA.
Vinitial=3.263705831-max(0.250,1.825mA*102.24ohm)
=3.013705831V; shared transition reserve=1uC.
thold=[641.52uF*(Vinitial-2.7V)-1uC]/2.642mA
=75.794309ms; conditional allocation.
Main C<=1mF. Rdis<=11.4211ohm, return/injection<=2.5mA;
10ms response +46.398757ms to90mV +10ms low hold
=66.398757ms total; hold margin=9.395552ms.
Recovery: raw>=3.70V, held>=3.25V; 1s settling PLUS
full source qualification; recharge model=0.657961s.
Qualify startup leakage, raw VIN=0, backfeed and cycling.
See ../design/sensor_alternative.md SENS-010;
../design/system_power_design.md SYS-007 for topology.

SYS-007 / PWR-006: SOURCE SUPERVISION / U11, R32-R34, C67-C71
Vtrip=Vref*(1+R32/R33)+Isense*R32.
Nominal=0.405*(1+732k/100k)=3.369600V.
Vref +/-2%, Isense +/-25nA; each R +/-0.1%, 25ppm/C, 100C.
Corner trip=3.263705831..3.476597632V.
C68: tau=(732k||100k)*1n=87.980769us (nominal).
C69-C71: 3x100n C0G; initial/temp=284.145..316.827nF.
CT delay=300/175+0.0005=1.714786s nominal.
0.6*(284.145/175+0.0005)=0.974511s MODEL, not guarantee.
Require qualified reset hold >=0.90s >650ms KILL blank +10ms.
R34=110k; ENhigh>=2.7-3.8uA*110k*1.01^2=2.273598V.
POR sink=(1.3-0.2)/(110k*0.99^2)+3.8uA=14.003041uA.
MAIN_PWR_EN drives raw U16 buffer; held Q3 clamps TPS63806 EN.
Held U12 also drives KILL/discharge; USB-clear still needed.
Current EN network and rail calculations: PWR-006.
Sources/Python: design/system_power_design.md SYS-007;
design/main_regulator_tps63806.md PWR-006. Reset coordination: #846.



Sheet: /Power button and source control/
File: power_button.kicad_sch

Title: RABP1 E-reader - power button and source control

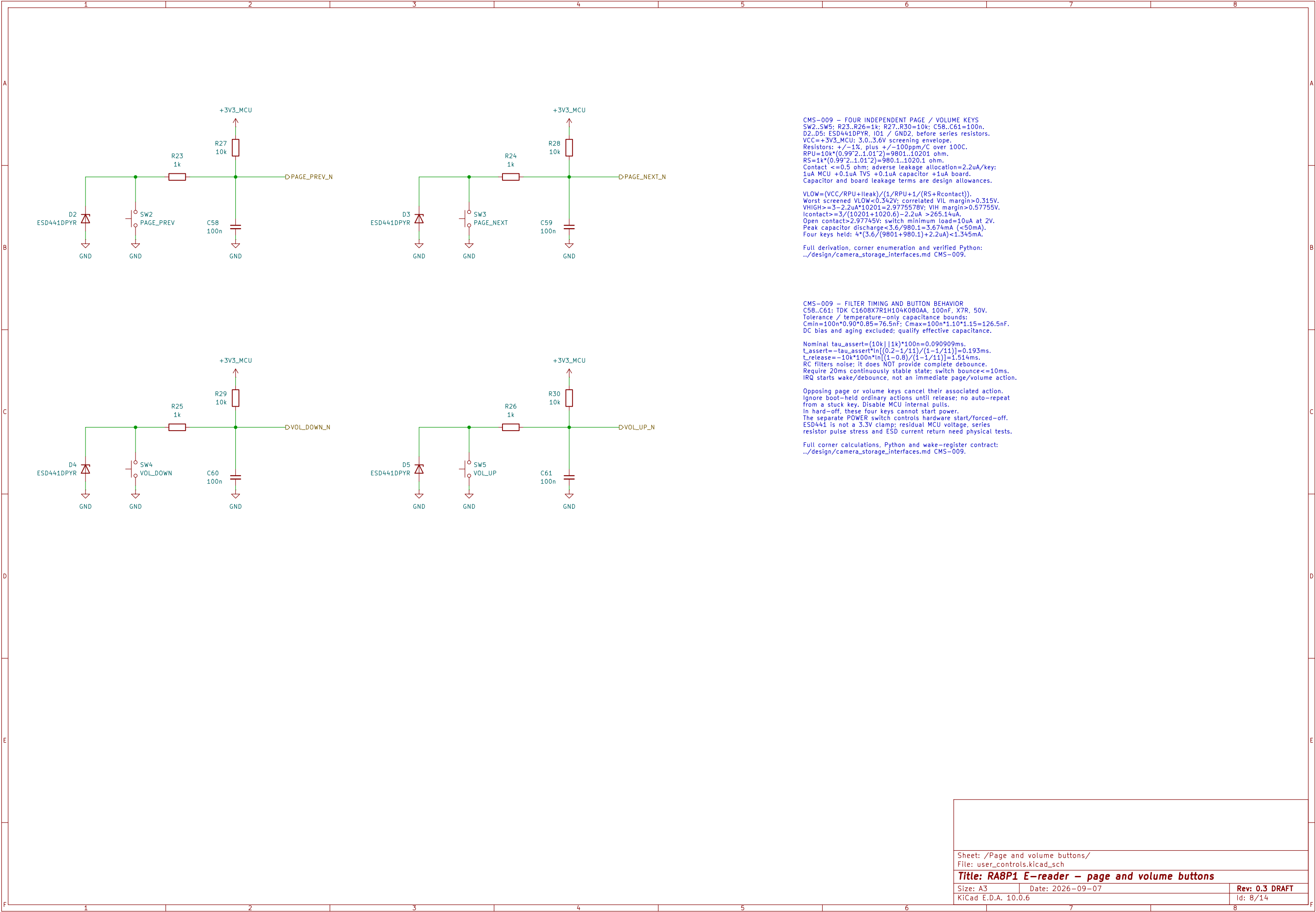
Size: A3

Date: 2026-09-07

Rev: 0.3 DRAFT

KiCad E.D.A. 10.0.6

Id: 7/14



VLOW=(VCC/RP/Ileak)/(1/RPU+1/(RS+Rcontact)).
Worst stressed VLOW<0.342V; correlated Vih margin>0.315V.
VHIGH>=3-2.2uA*10201=2.9775578V; Vih margin>0.57755V.
Icontact>=3/(10201+1020.6)-2.2uA >265.14uA.
Open contact=2.97745V: switch minimum load=10uA at 2V.
Peak capacitor discharge<3.6/9801=3.674mA (<50mA).
Four keys held: 4*(3.6/9801+9801)+2.2uA<1.345mA.

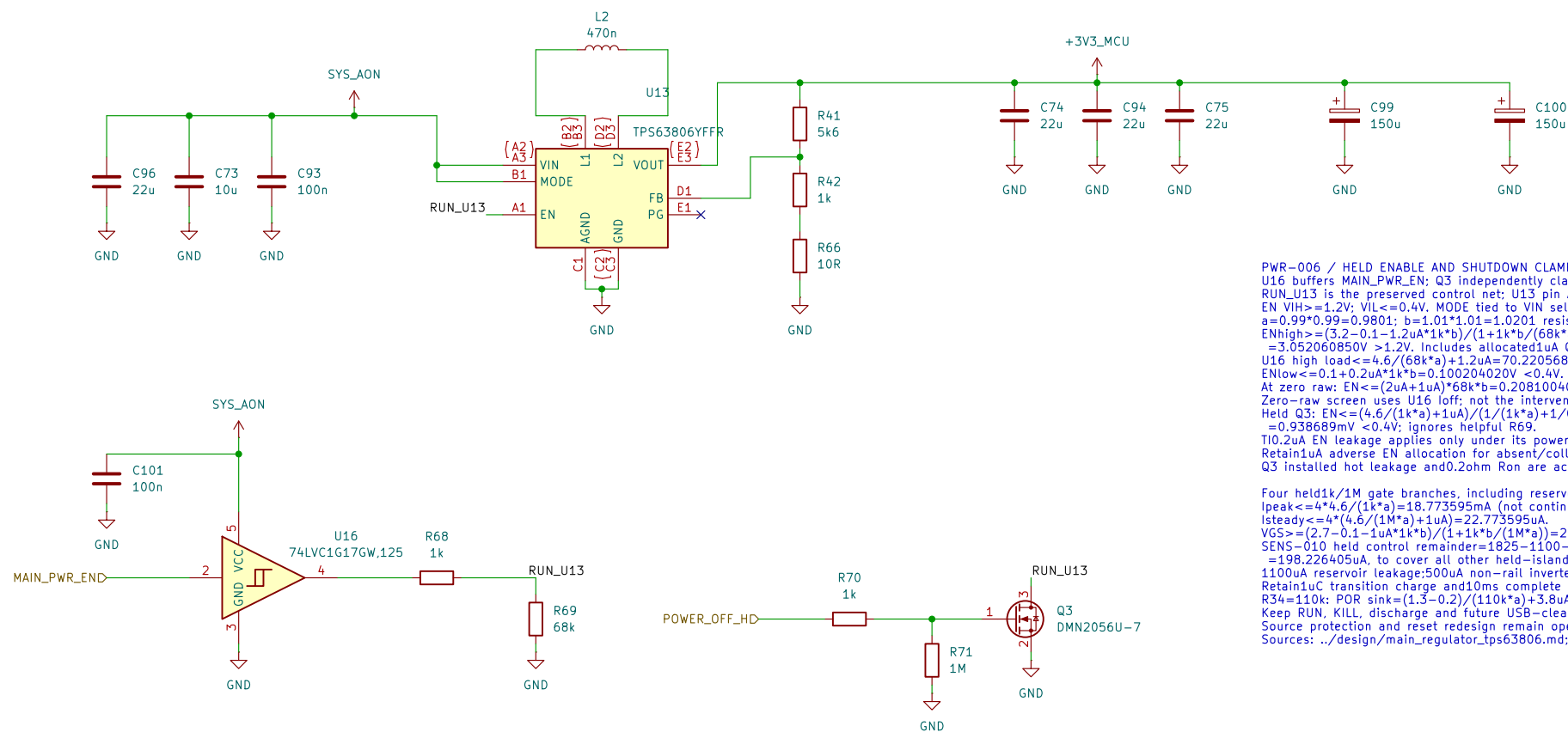
CMS-009 – FILTER TIMING AND BUTTON BEHAVIOR
C58..C61: TDK C1608X7R1H104K080AA, 100nF, X7R, 50V.
Tolerance / temperature-only capacitance bounds:
 $C_{min}=100n*0.90*0.85=76.5nF$; $C_{max}=100n*1.10*1.15=126.5nF$
DC bias and aging excluded; qualify effective capacitance.

Nominal tau_assert=(10k|1k)*100n=0.909090ms.
t_assert=-tau_assert*ln[(0.2-1/11)/(1-1/11)]=0.193ms.
t_release=-40k*100n*ln[(1-0.8)/(1-1/11)]=1.514ms.
RC filters noise; it does NOT provide complete debounce.
Require 20ms continuously stable state; switch bounce<=10ms.
IRQ starts wake/debounce, not an immediate page/volume action.

Opposing page or volume keys cancel their associated action.
Ignore boot-held ordinary actions until release; no auto-repeat
from a stuck key. Disable MCU internal pulls.
In hard-off, these four keys cannot start power.
The separate POWER switch controls hardware start/forced-off.
ESD441 is not a 3.3V clamp; residual MCU voltage, series
resistor pulse stress and ESD current return need physical tests.

Full corner calculations, Python and wake-register contract:
 ../design/camera_storage_interfaces.md CMS-009.

MAIN DIGITAL SUPPLY: TPS63806 / PWR-006
3.272277 V nominal; forced PWM; held shutdown clamp.
Divider and bypass calculations: design/main_regulator_tps63806.md
Reset coordination (#846), source protection and qualification remain open.
DEVELOPMENT SCHEMATIC – NOT FOR POWER-UP OR FABRICATION.



```

PWR-006 / HLD ENABLE AND SHUTDOWN CLAMP
U16 buffers MAIN_PWD_EN; Q3 independently clamps EN during raw collapse.
RUN_U13 is the preserved control net; U13 pin A1 is now EN.
EN VIH>=1.2V; VIL<=0.4V. MODE tied to VIN selects forced PWM.
=0.99+0.99*0.9801; b=1.01*1.01+1.0201 (resistor factors)
= (3.2-0.1-1.2uA)*1k*b/(1-0.99)
= 3.052060850uA. Included allclocks_U13 off leakage.
U16 high load = 4.6/(68k*1) + 1.2uA/70.220568uA <100uA.
ENlow = 0.1+0.2uA*1k*b = 0.100204020V <0.4V.
At zero raw: EN<=(2uA+1uA)*68k*b = 0.208100400V.
Zero-raw screen uses U16 loff; not the intervening undefined region.
Held Q3: EN<= (4.6/(1k*1)+1uA)/(1/(1k*1)+1/0.2ohm)
= 0.9386mV <0.4V; ignores held Q3.
T10.2uA EN leakage applies only under its powered test conditions.
RetainU1A adverse EN allocation for absent/collapsing raw.
Q3 installed hot leakage and0.2ohm raw are acceptance allocations.

```

```

Four held1k/1M gate branches, including reserved USB-clear:
Ipeak<=4*(4+6/(1k^a))=18.773595mA (not continuous ratings).
Isteady<=4*(4+6/(1M^a)+1A)=22.773595mA.
VG5=(2.7-0.1/(1A+1k^b))*(1+1k^b/(1M^a))=2.566278V.
SENS=high-level control, remainder=862.5mA=22.773595
=198.226450uA, to cover all other held-island loads.
1100uA reservoir leakage;500uA non-rail inverter-input allocation.
RetainLCU transition charge and10ms complete response allocation.
R34=110K; POR sink=(1.3-0.2)/(110K^a)+3.8uA=14.003041uA.
KILL=KILL pin, no charge and no current source separate.
Source protection and reset redesign remain open; no power-up approval.
Sources.../design/main_regulator_tps63806.m; SENS=010 revision.
```

PWR-Vom / FEEDBACK AND LOCAL PASSIVES
 $V_{nom} = 5.00 \times (1 + 5600 / (1000 + 10)) = 3.27227228 \text{ V}$
R41: 0.02%, 5ppm/C; R42: 0.01%, 2ppm/C; R66: 1%, 200ppm/C.
Installed boards: initial "TCR(1000)C"aging"assembly, then ohms.
 $R41 = 5600 \times (1 + (-0.0002)) \times (1 + (-0.0005)) \times (1 + (-0.001))$
 $\times (1 + (-0.0005)) \div (-0.02) = 5587.6629237, 5612.349243 \text{ ohm}$
 $R42 = 1000 \times (1 + (-0.0001)) \times (1 + (-0.0002)) \times (1 + (-0.001))$
 $\times (1 + (-0.0005)) \div (-0.01) = 998.190970, 1001.810970 \text{ ohm}$
 $R66 = 10 \times (1 + (-0.01)) \times (1 + (-0.02)) \times (1 + (-0.01)) \div (-0.05)$
 $= 9.954980, 10.455020 \text{ ohm}$ one combined aging/assembly 1%.
 $V_{in} = 5.00 \times (1 + 141680 / (R41 \times R42 \times R66)) = 5.00 \times (1 + 141680 / 1000000000)$
 $V_{in} = 5.005 \times (1 + 141680 / (R42 \times R66)) = 5.005 \times (1 + 141680 / 100000000)$
 $V_{static} = 3.226819680, 3.318012496 \text{ V (FPWM only)}$
One combined + / -75mV regulation/ripple/transient/routing allowance:
 $V_{complete} = 3.151819680, 3.393012496 \text{ V}$: qualification required.
SDRAM static VOH margin = $2.4 - 0.73 = 3.93012496 \text{ V} = 24.891253 \text{ mV}$.
This is not measured S/I/noise headroom or reset qualification.

12: 470nH +/-20%; catalog spread alone is insufficient.
Screen incoming0.412..0.518uH; allocated operating+/-10%
= +/-3708..0.5698uH versus T137..0.57uH. Quality hot/current.
C74/C75/C94: *322uF*0.8*0.85*0.6=6.282uF > T121uF
C96: 22uF*0.8*0.85*0.6=8.976uF > T14uF input minimum.
The capacitor combined resistance is not guaranteed.
Samsung typical output bias retention at3.393012496V-90.221554%.
Input bias, aging, mounted impedance and startup need verification.
Full assumptions, sources and executable Python: click this note.

PWR-006 / LDA, STORAGE AND SHUTDOWN
Allocation: 1.8A operating; 2.25A cold start screen, not measured load.
Pwr=3.393012496e+1,8=6.107424932E+3W
Thermal allocated Tjmax efficiency: Pin=84.33229991W; lin@3.2V=2,544759372A
Losses = 2.68078670W; loss = 0.678602499W
Tj90C efficiency: loss=0.678602499W
Tj90C=70C+78.8C/0.678602499W=123.474C.
2.5A board theta-JA is not final PCB thermal resistance.
Pin=3.393012496e+1,8=6.107424932E+3W
Converter limit, source limit, thermal and thermal envelope need validation.

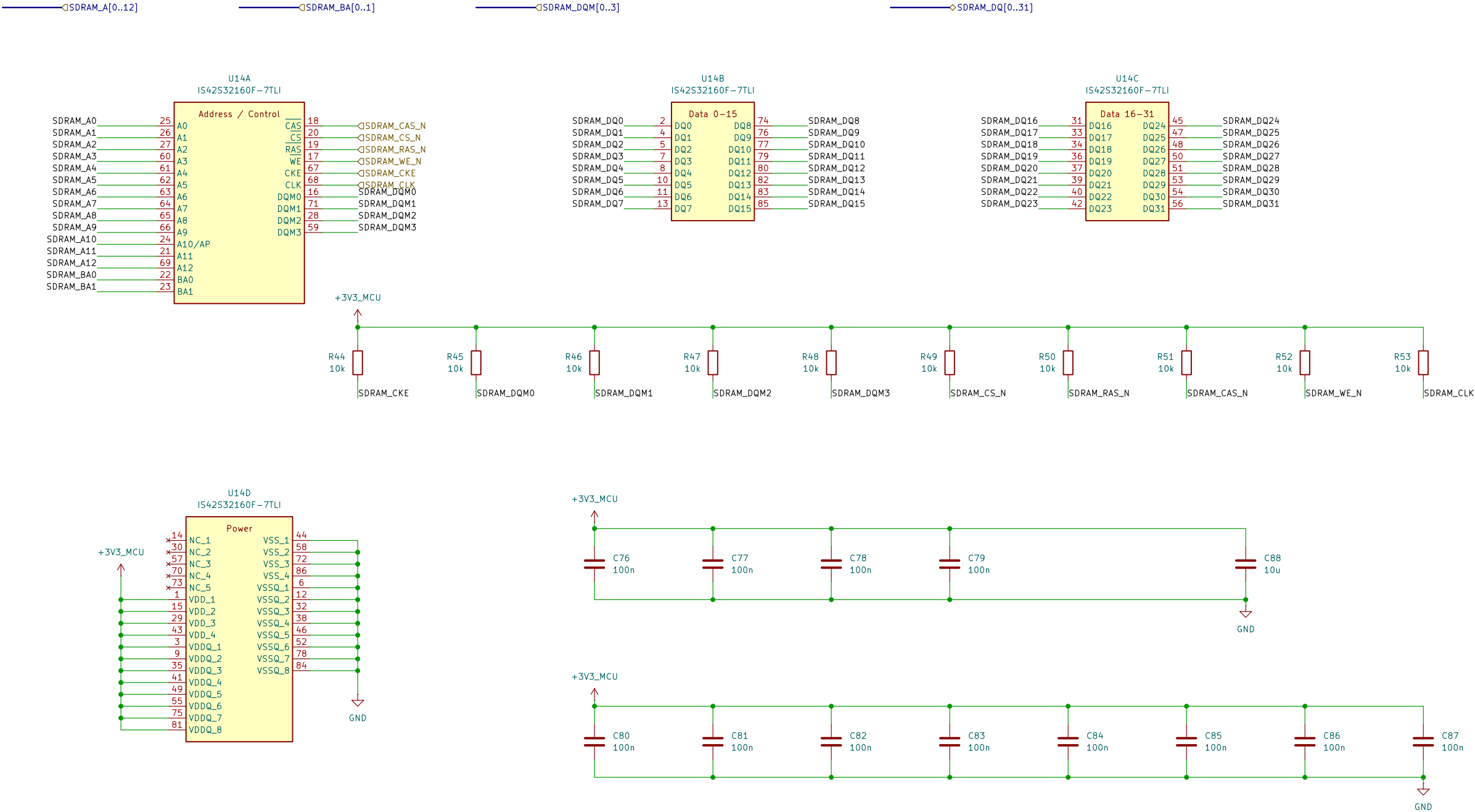
PWR=006 storage checkpoint (nominal, not effective minimum):
 Direct main469.21uF + C43 through FB1 10uF + key filters0.4uF
 =479.61uF cold; radio-on add±0.2uF =>489.81uF.
 C99/C100 remain±150uF each. C94 adds22uF to the output.
 Old bootstrap/private VCC/VF capacitors were not main-rail storage.
 Whole-main ceiling remains1mF; reconcile later peripheral additions.
 Cout lower acceptance and distributed PDN/startup are separate checks.

```

SENS-010 discharge revision: Rmax1+0.42110hm: return/injection<=+2.5mA.
Includes1.5mA key return +1mA other backfeed: Vinf=0.028552750V.
t=1mF+11.42110hm*ln((3.6-Vinf)/(0.09-Vinf))+10ms+10ms
=66.3987575ms total, below475.794309ms screen.
Includes1.0ms delay+10ms ramp/discharge+10ms delay+10ms below90mV.
Held budget:1.825mA control +0.817mA reverse;1uC transition reserve.
198.226405uA control remainder must cover other held-island loads.
SYS_AON source protection, loaded startup and fast faults remain open.
RSN_002 / RADIO-019: release margin remains conditional on rail/path bounds.
No more power or data for SENS-010 and correlation function are complete.
Full assumptions: ../design/main_regulator_tps63806.mt
shutdown revision/Python: ../design/sensor_alternative.md SENS-010.

```

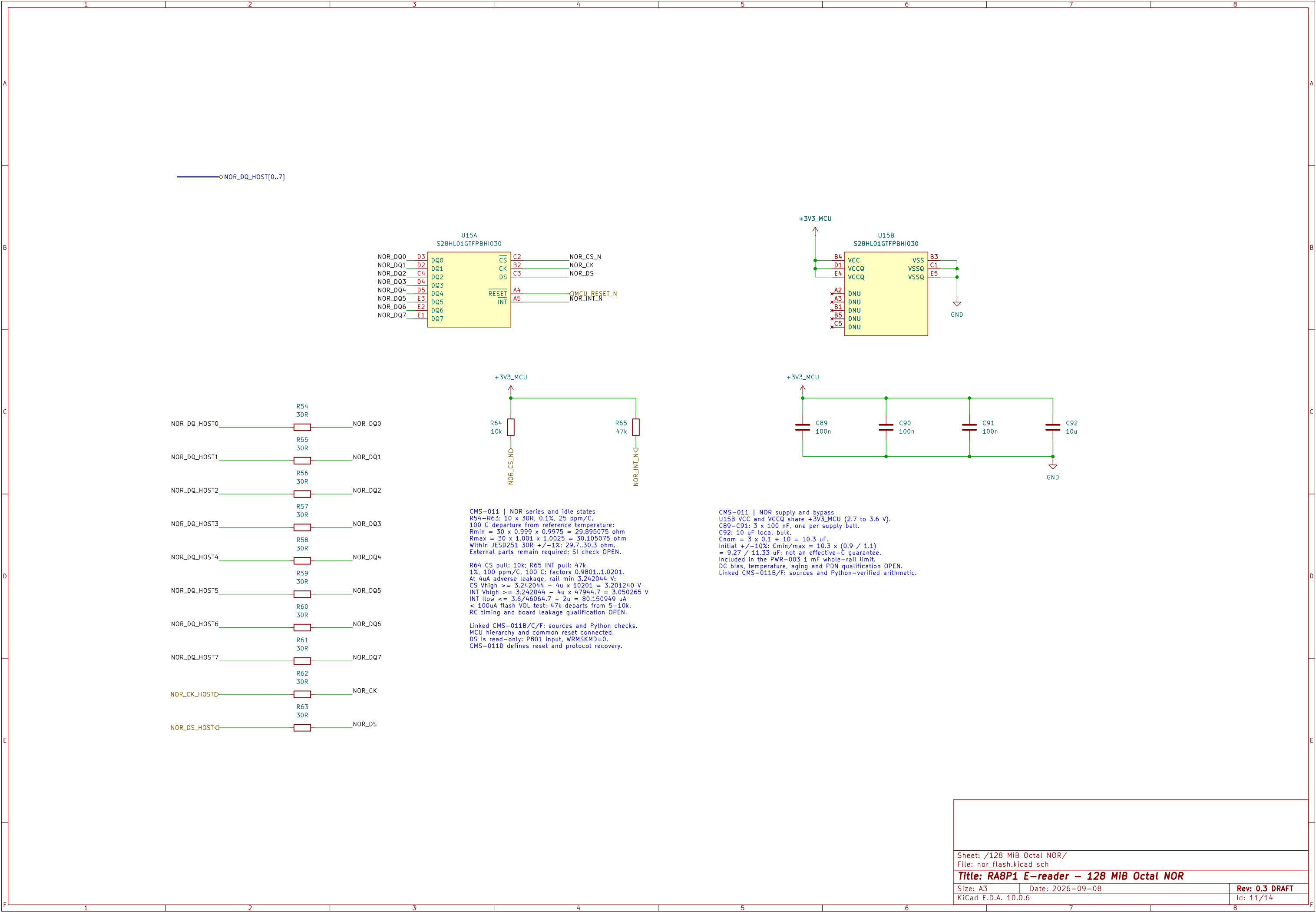
64 MiB SDRAM | ISSI IS42S32160F-7TLI
Signal and reset—default circuits | CMS-010.
Startup, SI/timing and hardware qualification remain open.

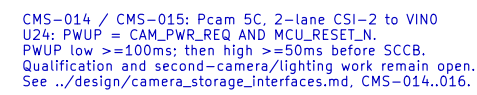


CMS-010 | SDRAM power and bypass
VDD and VDDQ share +3V3_MCU (3.0 to 3.6 V).
C76-C79: 4 x 100 nF for the four VDD pins.
C80-C87: 8 x 100 nF for the eight VDDQ pins.
C88: 10 uF local bulk; Cnom = 12 x 0.1 + 10 = 11.2 uF.
Included in the PWR-003 1 mF whole-rail limit.
Effective C, placement and PDN impedance require qualification.

CMS-010 | R44-R53: 10k reset defaults
Rail 3.0..3.6V: +/-1%, 100ppm/C. |dT|<=100C.
Rmin=10k*0.99*0.99=9801 ohm.
Rmax=10k*1.01*1.01=10201 ohm.
Ioff=5uA SDRAM+1uA MCU+1uA board=7uA.
VHIGHmin=3.0-7uA*10201=2.928593V (>2.0V VIH).
Board leakage and temperature screen require qualification.

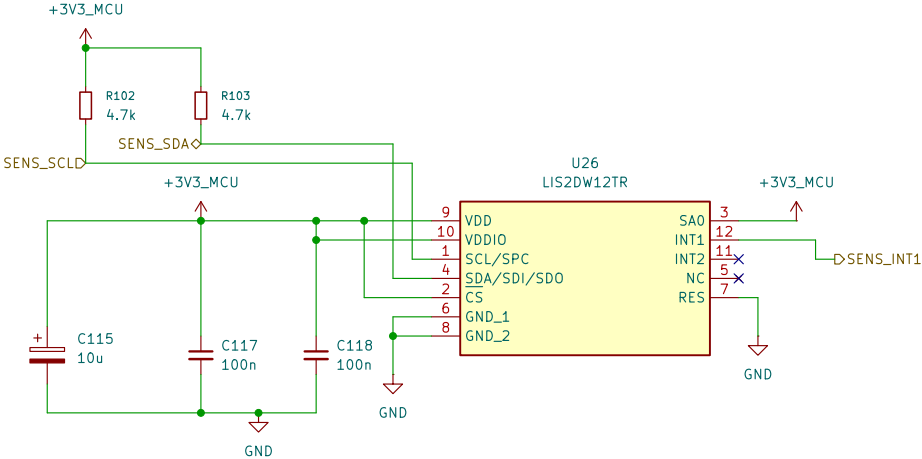
CMS-010 | Pull load and power states
Sink leakage: 5uA SDRAM+1uA board=6uA.
Isink=3.6/9801+6uA=0.373309mA (<0.374mA).
Pmax=3.6^2/9801=1.322314mW per pull.
All ten LOW: 10*3.6/9801=3.673095mA rail;
10*3.6^2/9801=13.223140mW resistor heat.
Reserve 4mA within the 250mA SDRAM allocation.
CKE/DQM and CS_N default HIGH; CLK HIGH is not a clock.
Switched-rail pulls; self-refresh actively holds CKE LOW.
Startup, dynamic I/O, SI and shutdown qualification OPEN.





Sheet: /Pcam 5C camera/ File: camera.kicad_sch		
Title:		
Size: A3	Date:	Rev:
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ORIENTATION SENSOR – WIP
U26: LIS2DW12TR; ST DS11811 Rev 9 pin map.
VDD pin 9 / VDDIO pin 10: +3V3_MCU; CS / SA0 high for I2C address 0x19.
C117/C118: local 100n bypass; GND pins 6/8 and RES pin 7 grounded.
C115: 10u/16V aluminum EEE–FM1C100R; place near VDD pin 9 with C117.
R102/R103: 4.7k SCL/SDA pull–ups; obsolete ADXL VREG network removed.
Host connected: SCL P410/K13; SDA P409/R16; INT1 P306/D11.
Wait 20ms after supply valid. Power–cycle below 100mV for >=10ms.
Pending: sourcing fields, pin modes, timing/power/passive qualification.
See design/sensor_alternative.md; circuit qualification remains open.



Sheet: /Orientation sensor/
File: orientation_sensor.kicad_sch

Title:

Size: A4
KiCad E.D.A. 10.0.6

Date:

Rev:

Id: 14/14